

Hamromaster

COMPLETE NOTES

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**For Hamromaster Online
Classes**

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Estimation

- Estimation can be defined as the statistical tool which is used to predict or estimate or guess the unknown value of population parameters with the help of sample evidence and past experience.
- There are two types of estimation as mentioned below
 - ▷ point Estimation.
 - ▷ interval Estimation.
- It is a inferential statistic.
- Sample की माध्यमवत् conclusion popⁿ मा दिने method की एक प्रकारकी Estimation. हो।
- The process in which we obtained the value of unknown population parameter with the help of sample data is estimation.
 - { नमूना data की माध्यमवत् / उपयोगमा हमीले अज्ञात popⁿ parameter की मूल्य / data etc प्राप्त गर्ने प्रक्रियालाई estimation भनिन्छ।
- Estimate: (अनुमान) : { Numerical value }
 - " An estimate is the numeric value of estimator. "
- Estimator (अनुमानक) : It is a rule, formula, or function that tells how to calculate an estimate.

(हामी कुन formula Used गर्दा तबलाई estimator भनिन्छ।)

- A point estimation is a single value prediction of a popⁿ parameter. for instance, a Sample mean is a point estimation of a population mean.
- An interval estimation is large values of prediction where the popⁿ parameter is expected to lie. Finding Confidence interval is the most common type of interval estimation.

Criteria for Good Estimators

- A good estimator is one which estimates the popⁿ parameter closest to the true value.
- The following are some of the criteria which should be satisfied by a good estimator.

C - consistency (तरक्थ हुनुपर्छ)

U - unbiasedness (fair हुनुपर्छ।)

S - sufficiency (सबैलाई सम्बोधन गर्दछ)

E - Efficiency (प्रयोग सिद्ध हुनु पर्छ।)

Numerical calculations

Confidence limit for Large Sample

- for mean
- for proportion

Sample Size determination

- for mean
- for proportion

How to identify for mean or proportion?

IS SD given?

yes

No

Mean Related

proportion Related



- Confidence Limit in Estimating population Mean for Large Sample Size

Following formula can be used to find the limit

$$\therefore \text{Limit} = \bar{X} \pm Z_{\alpha} \times S.E(\bar{X})$$

where,

$$S.E(\bar{X}) = \frac{\sigma}{\sqrt{n}} \text{ or } \frac{S}{\sqrt{n}}$$

for infinite popⁿ

* If we have finite popⁿ (i.e. N is given)

Case: 1: If sample is drawn with replacement (its like infinite)

$$S.E(\bar{X}) = \frac{\sigma}{\sqrt{n}} \text{ or } \frac{S}{\sqrt{n}}$$

case 2: If Sample is drawn, without replacement

$$SE(\bar{x}) = \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

OR

$$\frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Note: यदि (n/N) की मात्रा 0.05 भन्दा कम
भए $\sqrt{\frac{N-n}{N-1}}$ को add नगरिने हुन्छ।

palpali bharat

Q.1 The quality control manager at a light bulb factory needs to estimate the average life of a large shipment of light bulbs. The process standard deviation is known to be 100 hours. A random sample of 64 lights bulbs indicated a sample average life of 350 hours.

- (i) calculate the Standard error of them.
- (ii) Set up a 95% Confidence interval estimate of the average life of light bulb in this shipment.
- (iii) Set up a 99% confidence interval estimate of the true average life of light bulbs in this shipment assuming that

there are 1000 bulbs in consignment.

Solⁿ
Here,

[यहाँ n छोटी आकार का Sample
की वजह से popⁿ नहीं है]

Sample Size (n) = 64

Sample mean ($\bar{X}$) = 350 hours

Standard deviation (σ) = 100



Req. ① calculation of Standard Error

$$S.E(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{100}{\sqrt{64}} = 12.5$$

Req ① Limits at 95 %.

Confidence Level ($1-\alpha$) = 95%

Significance Level (α) = 5%

$$Z_{\alpha} = Z_{0.05} = 1.96 \rightarrow \text{for two tailed}$$

किन्ना ?

$\Rightarrow$ limit शनैपदि

Now,

$$\text{Limit} = \bar{X} \pm Z_{\alpha}(S.E)(\bar{X})$$

$$= 350 \pm 1.96 \times 12.5$$

$$= 350 \pm 24.5$$

$$\therefore \text{Lower Limit} = 350 - 24.5 = 325.5 \text{ hours}$$

$$\therefore \text{Upper Limit} = 350 + 24.5 = 374.5 \text{ hours.}$$

Q. (iii) Limits at 99%

$$\text{pop}^n \text{ Size } (N) = 1000$$

checking. Ratio

$$\frac{n}{N} = \frac{64}{1000} \Rightarrow 0.064$$

The ratio is greater than 0.05.. So
popⁿ size (N) should be adjusted
in the calculation of SE.

Now,

Confidence Level $(1-\alpha) = 99\%$

Significance Level $(\alpha) = 1\%$

$$Z_{\alpha} = Z_{0.01} = 2.58 \rightarrow \text{for two tailed}$$

Now,

$$\text{Limit} = \bar{X} \pm Z_{\alpha} \times \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

$$= 350 \pm 2.58 \times \frac{100}{\sqrt{64}} \sqrt{\frac{1000-64}{1000-1}}$$

$$= 350 \pm 2.58 \times 12.5 \times 0.968$$

$$= 350 \pm 31.22$$

$$\therefore \text{Upper Limit} = 350 + 31.22 = 381.22 \text{ hours}$$

$$\times \text{Lower Limit} = 350 - 31.22 = 318.78 \text{ hours}$$

Confidence Limit in Estimating population proportion for Large Sample size



$$\text{Limit} = p \pm z_{\alpha} \times \text{S.E.}(p)$$

where,

$$\text{S.E.}(p) = \sqrt{\frac{pq}{n}} \times \sqrt{\frac{N-n}{N-1}}$$

or

$$\text{S.E.}(p) = \sqrt{\frac{pq}{n}} \times \sqrt{\frac{N-n}{N}}$$

Note
 $p = \frac{x}{n}$ हूँ

यदि N दिखती है व
 (n/N) का Ratio 0.05
 वा सो बड़ा वही
 ह भवे मात्र यो रेखा
 भित्रको part धर्ने

- Q. In a large consignment there are 17500 apples from which 700 apples are taken as sample and 147 apples are found damaged. Find 96% confidence limits for the proportion of damaged units in the consignment.

Sol?

Here,

Sol

Population Size (N) = 17500

Sample Size (n) = 700

Number of damaged apple (X) = 147

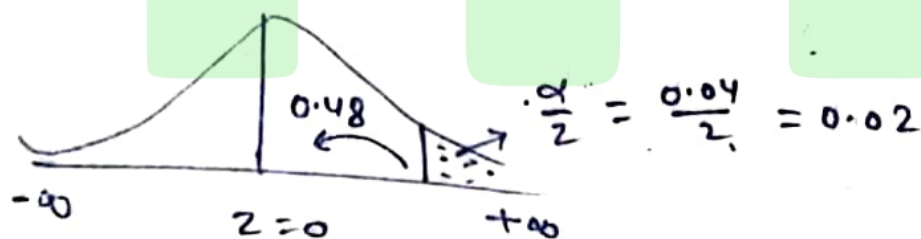
$$p = \frac{X}{n} = \frac{147}{700} = 0.21$$

$$q = 1 - p = 0.79$$

Checking Ratio : $\frac{n}{N} = \frac{700}{17500} = 0.04$

The ratio is less than 0.05, so popⁿ Size (N) should not be adjusted in the calculation of SE

$Z_{\alpha} = Z_{0.04}$ = Should be obtained from the area of normal Curve.



The prob. 0.48 approximately lies at $z = 0 + 0.05 = 2.05$ in the area of normal curve.

$$\therefore Z = 2.05$$

slow,

$$\text{Limit} = p \pm Z_{\alpha} \cdot S.E.(P)$$

$$= p \pm Z_{0.04} \sqrt{\frac{pq}{n-1}}$$



$$= p \pm Z_{0.04} \sqrt{\frac{0.21 \times 0.79}{700}}$$

$$= 0.21 \pm 0.0316$$

$$\text{Lower Limit} = 0.21 - 0.0316 \Rightarrow 0.1784$$

$$\text{Upper Limit} = 0.21 + 0.0316 \Rightarrow 0.2416$$

✓

Points to Remember:

(a) If reference for confidence level or significant level is not given in the question, then we consider one alternative from following 2 alternatives:

Alternative I: take $Z_{\alpha} = 1.96$ by assuming 5% significant level.

Alternative II: take $Z_{\alpha} = 3$ by assuming quality is almost assured.

(b) The difference between Sample Statistic and Popⁿ parameter (i.e. numerator part) is known as,

→ Error

→ Sampling Error

→ Errors of Estimate

→ denoted by 'E'

for mean	for proportion
Error (E) = $\bar{X} - \mu$	Error (E) = $p - P$

Determining Sample Size

Case I: If Mean Related.

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$$\text{or, } Z_{\alpha} = \frac{E}{\sigma / \sqrt{n}}$$

$$\text{or, } Z_{\alpha} = \frac{E}{\sigma} \times \sqrt{n}$$

$$\text{or, } \sqrt{n} = \frac{Z_{\alpha} \times \sigma}{E}$$

Sq. on both sides,

$$\text{or, } n = \left(\frac{Z_{\alpha} \sigma}{E} \right)^2$$

Case II: If proportion Related,

$$Z = \frac{p - P}{\sqrt{\frac{pq}{n}}}$$

$$\text{or } z_{\alpha} = \frac{E}{\sqrt{\frac{pq}{n}}}$$

$$\text{or } \sqrt{n} = \frac{z_{\alpha} \times \sqrt{pq}}{E}$$

$$\text{or } n = \left(\frac{z_{\alpha}}{E} \right)^2 pq$$

અહીં p & q તકરાર દેત અને 0.5 & 0.5 માટે



→ A researcher wants to estimate universe mean by using Sampling technique. what should be the Sample Size when the permissible error between parameter value and Sample Statistic in 95% of chance will not be more than 1.5 and popⁿ Standard deviation is 1.5.

Solⁿ

Here,

Sample Size (n) = ?

Error (E) = 1.5

S.D (σ) = 1.5

Confidence level (1-α) = 95%

Significant level (α) = 5%

∴ $z_{\alpha} = 1.96$ [Two tailed]

We know,

$$\begin{aligned}n &= \left(\frac{Z_{\alpha} \cdot \sigma}{E} \right)^2 \\&= \left(\frac{1.96 \times 15}{1.5} \right)^2 \\&= (19.6)^2 \\&= 384.16 \\&\approx 384.\end{aligned}$$

palpali bharat

- From a Sample of 400 computed mean & Standard Error of mean found to be 125 & 2.5 respectively. what Sample should be taken, to insure 99% Confidence that the popⁿ mean lies within more or less 5 of the Sample mean?

Solⁿ

Here,

Sample Size (n) = 400

Sample mean ($\bar{x}$) = 125

$$SE(\bar{x}) = \left(\frac{\sigma}{\sqrt{n}} \right) = 2.5$$

Confidence level $(1-\alpha) = 99\%$.

Significance $(\alpha) = 1\%$.
 $\therefore Z_{\alpha} = 2.576$ (Two tailed)

Error (E) = 5
Sample size (n) = ?

we have,

$$\frac{\sigma}{\sqrt{n}} = 2.5$$

$$\text{or, } \sigma = 2.5 \times \sqrt{400} = 50$$

Now,

$$n = \left(\frac{Z_{\alpha} \cdot \sigma}{E} \right)^2 = \left(\frac{2.576 \times 50}{5} \right)^2$$
$$= 663.58$$
$$\approx 664.$$



Trick :-

A manufacturing Company wishes to estimate the prob. of people who prefer their brand. It wishes to keep the error within 2 percent with a risk of 0.1336. How large a sample must be taken? Given that the chance of people prefer their brand are equally likely.

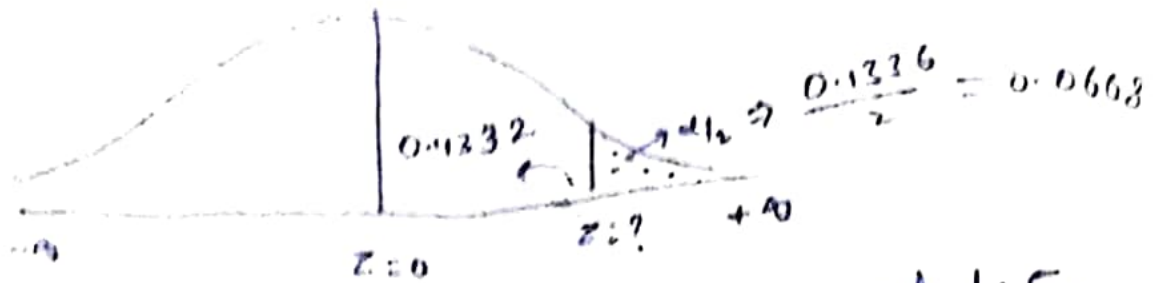
Solⁿ

Here,

$$\text{Error (E)} = 0.02$$

$$\text{Risk } (\alpha) = 0.1336$$

$$Z_{\alpha} = Z_{0.1336} \Rightarrow \left[\begin{array}{l} \text{obtained from the} \\ \text{area of normal curve} \end{array} \right]$$



The prob. 0.4332 exactly lies at 1.50
under the area of normal curve.

$$\therefore z_c = 1.50$$

proportion of brand preference (p) = 0.50

$$q = 1 - 0.50 \Rightarrow 0.50$$

Sample size (n) = ?

We know that

$$n = \left(\frac{z_\alpha}{E} \right)^2 p q$$

$$= \left(\frac{1.50}{0.02} \right)^2 \times 0.50 \times 0.50$$

$$= 1406.25$$

$$\approx 1406$$

Ex. 9.3

→ A Sample of 30 households drawn from a city area containing 14245 total households. It was known that the average household size = 3.4 and standard deviation = 1.22. Estimate the total city popⁿ at 95% Confidence level.

Solⁿ
Here

Total household (N) = 14845

Sample households (n) = 30

Average household size ($\bar{x}$) = 3.4

Total city popⁿ (Σx) = ? at 95% confidence level.

Standard deviation (S) = 1.22

Confidence interval at 95% confidence level is given by

$$= \bar{x} \pm z_{\alpha} \text{ S.E. } (\bar{x})$$

$$= 3.4 \pm 1.96 \cdot \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

$$= 3.4 \pm 1.96 \cdot \frac{1.22}{\sqrt{30}} \sqrt{\frac{14845-30}{14845-1}}$$

$$= 3.4 \pm 0.436$$

$$= \underbrace{2.964}_{\bar{x}_1}, \underbrace{3.836}_{\bar{x}_2}$$

Now, according to question,

Total popⁿ of the city = $(\bar{x}_1 N - \bar{x}_2 N)$

$$= (2.964 \times 14845 - 3.836 \times 14845)$$

$$= 4400.58 - 56945.42$$

$$\approx (4400 - 56945)$$

Notation :-

pop ⁿ character	sample character
(i) sample Size = (n)	Sample size = (n)
(ii) pop ⁿ mean = μ	Sample mean = $\bar{x}$
(iii) pop ⁿ S.d = σ	Sample S.d = s
(iv) pop ⁿ Variance = σ^2	Sample variance = s^2
(v) pop ⁿ proportion = p & q	Sample proportion = $\hat{p}$ and $\hat{q}$

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Exercise

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9.5 मा S.D $\Rightarrow \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$

(1, 2, 4, 5 मा $\rightarrow \bar{x} \pm z_{\alpha} \text{ S.E.}(\bar{x})$ लगाउने)

(6-10 मा) $\rightarrow p \pm z_{\alpha} \text{ S.E.}(p)$ लगाउने)

11 मा $p = \frac{x}{n}$ जस्तै

$\rightarrow \sqrt{\frac{pq}{n}}$

④ $\rightarrow \bar{x} \pm z_{\alpha} \text{ S.E.}(\bar{x}) \rightarrow \frac{s}{\sqrt{n}}$

⑥ $\rightarrow p \pm z_{\alpha} \text{ S.E.}(p) \rightarrow \sqrt{\frac{pq}{n}}$

12 $\rightarrow$ पदिको सँगै n find गर्ने हो।

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